

Defining priors in R; ‘rprior’

This prior allow the user to use a prior for θ in a form of an R-function which return $\log \pi(\theta)$ as a function of θ . The function needs **not** to vectorize. The prior function needs to be prepared using `inla.rprior.define()` before use (see below for an example).

Example

As an example, we implement the `loggamma`-prior, which for $\theta = \log(\text{precision})$ is

```
rprior.func = function(lprec) {  
  return (dgamma(exp(lprec), a, b, log = TRUE) + lprec)  
}
```

The last term is the log-Jacobian for the change of variable. Note that the prior-parameters, a and b , can be passed on when *preparing* the prior (a required step), using

```
rprior <- inla.rprior.define(rprior.func, a=1.0, b=0.1)
```

`rprior` can then be used as the argument `prior = rprior` in the `hyper` argument; see the following example.

```
rprior.func = function(lprec) {  
  return (dgamma(exp(lprec), a, b, log = TRUE) + lprec)  
}
```

```
rprior <- inla.rprior.define(rprior.func, a = 1, b = 0.1)
```

```
prior.expression = "expression:  
  a = 1;  
  b = 0.1;  
  precision = exp(lprec);  
  logdens = log(b^a) - lgamma(a)  
             + (a-1)*lprec - b*precision;  
  ljacobian = lprec;  
  return(logdens + ljacobian);"
```

```
prior.func = function(lprec) {  
  a = 1; b = 0.1;  
  return (dgamma(exp(lprec), a, b, log = TRUE) + lprec)  
}  
lprec = seq(-10, 10, len=1000)  
prior.table = paste(c("table:", cbind(lprec, prior.func(lprec))),  
  sep = "", collapse = " ")
```

```
n = 100  
y = rnorm(n)
```

```
r = inla(y~1,  
  data = data.frame(y),  
  control.family = list(  
    hyper = list(  
      prec = list(  
        prior = "loggamma",
```

```

param = c(1, 0.1))))))

rr = inla(y~1,
  data = data.frame(y),
  control.family = list(
    hyper = list(
      prec = list(
        prior = prior.expression))))

rrr = inla(y~1,
  data = data.frame(y),
  control.family = list(
    hyper = list(
      prec = list(
        prior = prior.table))))

rrrr = inla(y~1,
  data = data.frame(y),
  control.family = list(
    hyper = list(
      prec = list(
        prior = rprior))))

round(c(r$mlik[1], rr$mlik[1], rrr$mlik[1], rrrr$mlik[1]), 5)

```